MTH-645 Optimization Techniques
Assignment (1)

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It is required to find the minimum of the unimodal function $F(x, y) = 15e^{-x} + 2e^{y}$ along the segment joining the two points $(1, 1)$ and $(5, 4)$ within an interval of length 0.1. Use both Fibonacci search and Golden section methods to locate the minimum.

Solution:

a. Fibonacci search

$$
\begin{aligned}
L_0 &= \sqrt{(5-1)^2 + (4-1)^2} = \sqrt{25} = 5. \\
L_n &= 0.1. \\
reduction(r) &= L_0/L_n = 5/0.1 = 50. \\
N_{fib} &\geq r = 50 \implies N_{fib} = 55. \\
\text{Fibonacci Sequence} &: 1, 1, 2, 3, 5, 8, 13, 21, 34, 55 \\
iterations &= 8.
\end{aligned}
$$

The line segment between a and b is given by the vector:

$$L_0 = (5-1, 4, 1) = (4, 3).$$

x_1 and x_2 are given by the equations:

$$
\begin{aligned}
x_1 &= a + \frac{F_{n-2}}{F_n} L_0 \\
x_2 &= b - \frac{F_{n-2}}{F_n} L_0 = a + \frac{F_{n-1}}{F_n} L_0.
\end{aligned}
$$

Iterations:

iteration	a	x_1	x_2	b	$f(x_1)$	comp	$f(x_2)$
1	(1,1)	(2.53, 2.15)	(3.47, 2.85)	(5,4)	18.36	$<$	35.04
2	(1,1)	(1.94, 1.70)	(2.53, 2.15)	(3.47, 2.85)	13.1	$<$	18.36
3	(1,1)	(1.59, 1.44)	(1.94, 1.70)	(2.53, 2.15)	11.5	$<$	13.1
4	(1,1)	(1.35, 1.26)	(1.59, 1.44)	(1.94, 1.70)	10.94	$<$	11.5
5	(1,1)	(1.24, 1.18)	(1.35, 1.26)	(1.59, 1.44)	10.85	$<$	10.94
6	(1,1)	(1.10, 1.08)	(1.24, 1.18)	(1.35, 1.26)	10.88	$>$	10.85
7	(1.10, 1.08)	(1.24, 1.18)	(1.21, 1.16)	(1.35, 1.26)	10.85	$=$	10.85

$$x^* = midpoint(a, b) = (1.23, 1.13)$$

b. Golden Section

First, let's compute the required number of iterations:

$$
\begin{aligned}
&(0.618)^n = \frac{1}{50} \\
&\implies n \ln(0.618) = \ln(\frac{1}{50}) \\
&\implies n = 8.13 \\
&\implies n = 9.
\end{aligned}
$$

x_1 and x_2 are given by the equations:

$$
\begin{aligned}
x_1 &= a + 0.618 L_0 \\
x_2 &= b - 0.618 L_0
\end{aligned}
$$

Iterations:

iteration	a	x_1	x_2	b	$f(x_1)$	comp	$f(x_2)$
1	(1.00, 1.00)	(2.52, 2.14)	(3.47, 2.85)	(5.00, 4.00)	18.29	$<$	35.18
2	(1.00, 1.00)	(1.94, 1.70)	(2.52, 2.14)	(3.47, 2.85)	13.18	$<$	18.29
3	(1.00, 1.00)	(1.58, 1.43)	(1.94, 1.70)	(2.52, 2.14)	11.50	$<$	13.18
4	(1.00, 1.00)	(1.36, 1.27)	(1.58, 1.43)	(1.94, 1.70)	10.97	$<$	11.50
5	(1.00, 1.00)	(1.22, 1.16)	(1.36, 1.27)	(1.58, 1.43)	10.84	$<$	10.97
6	(1.00, 1.00)	(1.13, 1.10)	(1.22, 1.16)	(1.36, 1.27)	10.83	$<$	10.84
7	(1.00, 1.00)	(1.08, 1.06)	(1.13, 1.10)	(1.22, 1.16)	10.86	$>$	10.83
8	(1.08, 1.06)	(1.13, 1.10)	(1.17, 1.12)	(1.22, 1.16)	10.83	$=$	10.83

$$x^* = midpoint(a, b) = (1.15, 1.11)$$

2.

It is required to find the maximum of the unimodal function $F = \ln x - 2x^2$ within the interval $(0, 1)$. Use the method which requires the minimum number of function evaluations to locate the maximum in an interval of length 0.05.

Solution:

We transform the problem into a minimization problem by multiply by -1, we get: the new objective function:

$$G(x) = 2x^2 - \ln x$$

The **Fibonacci search** method is the method that requires the minimum number of function evaluations. So we use it for the solution.

$$\begin{aligned} L_0 &= 1. \\ L_n &= 0.05. \\ reduction(r) &= L_0/L_n = 1/0.05 = 20. \\ N_{fib} &\geq r = 20 \implies N_{fib} = 21. \\ \text{Fibonacci Sequence} &: 1, 1, 2, 3, 5, 8, 13, 21 \\ iterations &= 6. \end{aligned}$$

Iterations:

iteration	a	x_1	x_2	b	$f(x_1)$	comp	$f(x_2)$
1	0.00	0.38	0.61	1.00	1.25	$>$	1.24
2	0.38	0.61	0.76	1.00	1.24	$<$	1.43
3	0.38	0.52	0.61	0.76	1.19	$<$	1.24
4	0.38	0.47	0.52	0.61	1.19	$>$	1.19
5	0.47	0.52	0.57	0.61	1.19	$<$	1.21
6	0.47	0.50	0.53	0.56	1.19	$<$	1.20

$$x^* = midpoint(a, b) = 0.52$$

3.

a. Use Fibonacci search to find the point of minimum energy $E = 14x^2 + y^2 + z^2 - 4xy - 6xz - 3x$ along the segment joining the two points $(1, 2, 4)$ and $(3, 5, 8)$ with an error $\epsilon \leq 0.05$.

b. Comment on the relation between the gradient at the point of minimum and the segment direction.

Solution:

a.

$$\begin{aligned}
\vec{L_0} &= (3-1, 5-2, 8-4) = (2, 3, 4). \\
L_0 &= \sqrt{(3-1)^2 + (5-2)^2 + (8-4)^2} = 5.39. \\
L_n &= 0.05. \\
reduction(r) &= L_0/L_n = 5.39/0.05 = 107.8. \\
N_{fib} \geq r &= 107.8 \implies N_{fib} = 144. \\
\text{Fibonacci Sequence} &: 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144 \\
iterations &= 10.
\end{aligned}$$

Iterations:

iteration	a	x_1	x_2	b	$f(x_1)$	comp	$f(x_2)$
1	(1.00, 2.00, 4.00)	(1.76, 3.14, 5.52)	(2.23, 3.85, 6.47)	(3.00, 5.00, 8.00)	-1.97	$<$	-1.27
2	(1.00, 2.00, 4.00)	(1.47, 2.70, 4.94)	(1.76, 3.14, 5.52)	(2.23, 3.85, 6.47)	-1.91	$>$	-1.97
3	(1.47, 2.70, 4.94)	(1.76, 3.14, 5.52)	(1.94, 3.41, 5.88)	(2.23, 3.85, 6.47)	-1.97	$<$	-1.82
4	(1.47, 2.70, 4.94)	(1.65, 2.97, 5.30)	(1.76, 3.14, 5.52)	(1.94, 3.41, 5.88)	-1.99	$<$	-1.97
5	(1.47, 2.70, 4.94)	(1.58, 2.87, 5.16)	(1.65, 2.97, 5.30)	(1.76, 3.14, 5.52)	-1.98	$>$	-1.99
6	(1.58, 2.87, 5.16)	(1.65, 2.97, 5.30)	(1.69, 3.04, 5.39)	(1.76, 3.14, 5.52)	-1.99	$<$	-1.99
7	(1.58, 2.87, 5.16)	(1.62, 2.93, 5.25)	(1.65, 2.97, 5.30)	(1.69, 3.04, 5.39)	-1.99	$>$	-1.99
8	(1.62, 2.93, 5.25)	(1.65, 2.97, 5.30)	(1.66, 3.00, 5.33)	(1.69, 3.04, 5.39)	-1.99	$>$	-1.99
9	(1.65, 2.97, 5.30)	(1.66, 3.00, 5.33)	(1.67, 3.01, 5.35)	(1.69, 3.04, 5.39)	-1.99	$<$	-1.99
x	(1.65, 2.97, 5.30)	(1.66, 2.99, 5.32)	(1.66, 3.00, 5.33)	(1.67, 3.01, 5.35)	-1.99	$>$	-1.99

$$x^* = midpoint(a, b) = (1.66, 2.99, 5.33)$$

b.

$$\vec{\nabla}(E) = \left(\frac{\partial E}{\partial x}, \frac{\partial E}{\partial y}, \frac{\partial E}{\partial z}\right) = (28x - 4y - 6z - 3, 2y - 4x, 2z - 6x)$$

$$\nabla(E(1.66, 2.99, 5.33)) = (-0.46, -0.66, 0.7)$$

Let θ be the angle between the segment direction, $\vec{L_0}$, and the gradient direction at the point of minimum, $\vec{\nabla}(E(1.66, 2.99, 5.33))$. Then:

$$\begin{aligned}
\cos(\theta) &= \frac{u \cdot v}{\|u\|\|v\|} \\
&= \frac{(2, 3, 4) \cdot (-0.46, -0.66, 0.7)}{5.39 * 3.02} \\
&= \frac{2 * (-0.46) + 3 * (-0.66) + 4 * 0.7}{5.39 * 1.07} = 0.03 \\
\implies \theta &= 88.
\end{aligned}$$

This shows that the directions nearly orthogonal.

4.

Consider the problem: Minimize $(x+2y-4)^2+(3x-4y-2)^2$. Starting at $(0,0)$, find the minimum along the steepest decent direction at $(0,0)$, find the minimum along the steepest decent direction at (0, 0) using Powell's quadratic interpolation and $\delta = 1$. How many quadratic interpolations are expected to lead to the minimum in this direction?

Solution:

Denote $F(x,y) = (x+2y-4)^2+(3x-4y-2)^2$.
First, we compute the gradient:

$$\nabla(F(x,y) =) \left(\frac{\partial E}{\partial x}, \frac{\partial E}{\partial y}\right)$$

$$\frac{\partial E}{\partial x} = (2(x+2y-4)(1)+2(3x-4y-2)(3))$$

$$\frac{\partial E}{\partial y} = 2(x+2y-4)(2)+2(3x-4y-2)(-4)$$

$$\implies \nabla(F(0,0)) = (-20,0)$$

Next, we compute the steepest descent direction at $(0,0)$:

$$-\nabla(F(0,0)) = (20,0)$$

$$\vec{d} = normalize\left((20,0)\right) = (1,0)\,.$$

Along the steepest descent direction, we parameterize (x,y) in terms of Δ:

$$(x,y) = (0,0) + \Delta\,(1,0) = (\Delta, 0)$$

And parametrize F in terms of Δ:

$$F(\Delta) = F(\Delta,0) = (\Delta-4)^2+(3\Delta-2)^2$$
$$= 10\Delta^2 - 20\Delta + 20.$$

Since only the initial guess point is given without a search interval, our strategy will be to take consecutive δ steps along the steepest descent direction until we find an appropriate search interval:

Δ	$F(\Delta)$
0	20
1	10
2	20

Since $F(\Delta) < F(0)$ and $F(\Delta) < F(2\Delta)$, then $[0, 2\Delta]$ is an appropriate search interval.
Next, we use Powell's quadratic interpolation at the 3 points: 0, Δ, 2Δ:

$$x_c = \frac{a+b}{2} = \Delta$$
$$y = x - x_c$$
$$G(y) = ay^2 + by + c$$
$$G(0) = c = F(\Delta) = 10$$
$$G(\delta) = G^+ = 20$$
$$G(-\delta) = G^- = 20$$

(To simplify the calculation, I'll just look up the formula for y^* from the slides):

$$y^* = -\frac{G^+ - G^-}{G^+ + G^- - 2G^0}\frac{\Delta}{2}$$
$$= -\frac{(20-20)}{20+20-2*10}\frac{1}{2} = 0.$$

$$x^* = y^* + x_c$$
$$= 0 + \Delta = 1.$$

We don't need to perform further searches, since the current interval length $= 2\delta \implies \epsilon = \frac{2\delta}{2} = \delta$, achieves the required accuracy.

5.

Construct three simplices of the simplex method to find the minimum of $F = 5x^2 - 4xy + 10y^2 - 7x$ using the initial simplex with vertices: (0,0), (0.5, 0), (0, 0.5).
Show the steps on a graph paper.

Solution:

Assume an expansion coefficient $\gamma = 2$ and a contraction coefficient $\beta = 1/2$.

Iteration 1:

Step 1: Evaluate the function at all vertiex points:

$$x_1 = (0,0) \qquad F(x_1) = 0$$
$$x_2 = (0.5,0) \qquad F(x_2) = -2.25$$
$$x_3 = (0,0.5) \qquad F(x_3) = 2.5$$

$\implies x_l = x_2, \quad x_h = x_3.$

Step 2: Find the centroix point x_0:

$$x_0 = \frac{1}{2}(x_1 + x_2) = (0.25, 0)$$
$$F(x_0) = -1.44.$$

Step 3: Find the reflection point x_r:

$$x_r = 2x_0 - x_h = (0.5, -0.5)$$
$$F(x_r) = 1.25.$$

Step 4: Decision point (accept OR expand OR contract):

$F(x_r) < F(x_h)$, but $F(x_r)$ is larger than all remaining points $\implies$ attemp contraction.

Step 5: Attempt contraction:

$$x_c = \beta x_h + (1-\beta)x_0 = (0.125, 0.25)$$
$$F(x_c) = -0.3.$$

$F(x_c) < \min[F(x_h), F(x_r)] \implies$ accept x_c.

Step 6: Test for convergence (standard-deviation):

$$Q = \sqrt{\frac{\sum_{i=1}^{n}[F(x_i) - F(x_0)]^2}{n}} = 1.16.$$

Iteration 2:

Step 1: Evaluate the function at all vertiex points:

$$x_1 = (0,0) \qquad F(x_1) = 0$$
$$x_2 = (0.5,0) \qquad F(x_2) = -2.25$$
$$x_3 = (0.125,0.25) \qquad F(x_3) = -0.3$$

$\implies x_l = x_2, \quad x_h = x_1.$

Step 2: Find the centroix point x_0:

$$
\begin{aligned}
x_0 &= \frac{1}{2}(x_2 + x_3) = (0.31, 0.125) \\
F(x_0) &= -1.69.
\end{aligned}
$$

Step 3: Find the reflection point x_r:

$$
\begin{aligned}
x_r &= 2x_0 - x_h = (0.62, 0.25) \\
F(x_r) &= -2.41
\end{aligned}
$$

Step 4: Decision point (accept OR expand OR contract):

$F(x_r)$ is less than all other points $\implies$ attemp expansion.

Step 5: Attempt expansion:

$$
\begin{aligned}
x_e &= 2x_r - x_0 = (0.93, 0.375) \\
F(x_e) &= -2.17
\end{aligned}
$$

$F(x_c) > F(x_r) \implies$ reject x_e and accept x_r.

Step 6: Test for convergence (standard-deviation):

$$
Q = \sqrt{\frac{\sum_{i=1}^{n}[F(x_i) - F(x_0)]^2}{n}} = 0.96.
$$

Iteration 3:

Step 1: Evaluate the function at all vertiex points:

$$
\begin{aligned}
x_1 &= (0.62, 0.25) & F(x_1) &= -2.41 \\
x_2 &= (0.5, 0) & F(x_2) &= -2.25 \\
x_3 &= (0.125, 0.25) & F(x_3) &= -0.3
\end{aligned}
$$

$\implies x_l = x_1, \quad x_h = x_3.$

Step 2: Find the centroix point x_0:

$$
\begin{aligned}
x_0 &= \frac{1}{2}(x_1 + x_1) = (0.56, 0.125) \\
F(x_0) &= -2.48.
\end{aligned}
$$

Step 3: Find the reflection point x_r:

$$
\begin{aligned}
x_r &= 2x_0 - x_h = (0.995, 0) \\
F(x_r) &= -2.01
\end{aligned}
$$

Step 4: Decision point (accept OR expand OR contract):

$F(x_r) < F(x_h)$, but $F(x_r)$ is larger than all remaining points $\implies$ attemp contraction.

Step 5: Attempt contraction:

$$\begin{aligned} x_c &= \beta x_h + (1-\beta)x_0 = (0.34, 0.19) \\ F(x_c) &= -1.7. \end{aligned}$$

$F(x_c) > F(x_r) \implies$ reject x_c and accept x_r.

Step 6: Test for convergence (standard-deviation):

$$Q = \sqrt{\frac{\sum_{i=1}^{n}[F(x_i) - F(x_0)]^2}{n}} = 0.3.$$

The final simplex is given by:

$$\begin{aligned} x_1 &= (0.62, 0.25) & F(x_1) &= -2.41 \\ x_2 &= (0.5, 0) & F(x_2) &= -2.25 \\ x_3 &= (0.995, 0) & F(x_3) &= -2.01. \end{aligned}$$

and the solution is given by:

$$\begin{aligned} x_0 &= \frac{1}{2}(x_1 + x_1) = (0.56, 0.125) \\ F(x_0) &= -2.48. \end{aligned}$$

6.

Carry out three steps to find three new base points of pattern search to Minimize

$$F = x_1^2 x_2^2 + (x_1 - 2)^2 (x_2 - 3)^2$$

starting at $x = (1, 1), \delta = 0.2$ for both directions. Show the steps taken on graph paper.

Solution:

Search directions (d_i):

$$\Big((1,0), (-1,0), (0,1), (0,-1)\Big)$$

First search:

Step 1: Compute base point:

$$b^0 = (1,1)$$
$$F(b^0) = 5.$$

Step 2: Make exploratory moves:

direction	b^0	$F(b^0)$	x^1	$F(x^1)$	$F(x^1)$ comp $f(b^0)$	ACCEPTED?
d_1	$(1,1)$	5	$(1.2,1)$	4	$<$	YES

Step 3: Make pattern move:

$$x^p = b^0 + 2\delta d_1 = (1.4, 1)$$
$$F(x^p) = 3.4.$$

direction	b^0	$F(b^0)$	x^1	$F(x^1)$	$F(x^1)$ comp $f(b^0)$	ACCEPTED?
d_1	$(1,1)$	5	$(1.6,1)$	3.2	$<$	YES

$\implies b^1 = (1.6, 1).$

Second search:

Step 1: Compute base point:

$$b^1 = (1.6,1)$$
$$F(b^1) = 3.2.$$

Step 2: Make exploratory moves:

direction	b^1	$F(b^1)$	x^2	$F(x^2)$	$F(x^2)$ comp $f(b^1)$	ACCEPTED?
d_1	$(1.6,1)$	3.4	$(1.8,1)$	3.4	$>$	NO
d_2	$(1.6,1)$	3.4	$(1.4,1)$	3.4	$>$	NO
d_3	$(1.6,1)$	3.4	$(1.6,1.2)$	4.2	$>$	NO
d_4	$(1.6,1)$	3.4	$(1.6,0.8)$	2.4	$<$	YES

Step 3: Make pattern move:

$$x^p = b^1 + 2\delta d_4 = (1.6, 0.6)$$
$$F(x^p) = 1.8.$$

direction	b^1	$F(b^1)$	x^2	$F(x^2)$	$F(x^2)$ comp $f(b^1)$	ACCEPTED?
d_1	$(1.6,1)$	5	$(1.6,0.4)$	1.5	$<$	YES

$\implies b^2 = (1.6, 0.4).$

Third search:

Step 1: Compute base point:

$$b^2 = (1.6, 0.4)$$
$$F(b^2) = 1.5.$$

Step 2: Make exploratory moves:

direction	b^2	$F(b^2)$	x^3	$F(x^3)$	$F(x^3)$ comp $f(b^2)$	ACCEPTED?
d_1	$(1.6, 0.4)$	1.5	$(1.8, 0.4)$	0.79	$<$	ACCEPTED

Step 3: Make pattern move:

$$x^p = b^2 + 2\delta d_1 = (2, 0.4)$$
$$F(x^p) = 0.64.$$

direction	b^2	$F(b^2)$	x^3	$F(x^3)$	$F(x^3)$ comp $f(b^2)$	ACCEPTED?
d_1	$(1.6, 0.4)$	1.5	$(2.2, 0.4)$	1	$<$	YES

$\implies b^3 = (2.2, 0.4)$.